

Coding & Solution

Date	13 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters – AI-Based Flood Prediction System
Maximum Marks	5 Marks

Coding & Solution:

The **Rising Waters** project implements an AI-powered flood prediction system using Python, Flask, and the XGBoost machine learning algorithm. The application processes historical flood records and real-time weather parameters to predict flood risks and generate early warning alerts. The solution follows modular programming practices, ensuring maintainability, scalability, and ease of deployment.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/238x1a0414-eng/Rising-Waters
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap
Key Features Implemented	• Historical flood data preprocessing • Weather data analysis • Feature engineering and scaling

	• XGBoost-based flood prediction • Real-time flood risk prediction • Early warning alert generation • Interactive web dashboard • Prediction history and reporting
Pending / Incomplete Features	• Live weather API integration (optional future enhancement) • SMS/Email notification service • Mobile application support
Setup / Run Instructions	1. Install Python and required libraries using pip install -r requirements.txt. 2. Place floods.save, transform.save, and the dataset in the project directory. 3. Run the application using python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter weather parameters to obtain flood risk predictions.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

1. The project follows a modular architecture with separate components for preprocessing, prediction, alert generation, and web interface.
2. The XGBoost model is trained using historical flood data to achieve high prediction accuracy.
3. Flask serves as the backend framework, enabling smooth integration between the machine learning model and the user interface.
4. The solution is designed for scalability and can be enhanced with live weather APIs, cloud deployment, and mobile notifications in future versions.

Outcome:

The Rising Waters system successfully integrates machine learning with a web-based application to deliver accurate flood predictions and early warning alerts. The solution demonstrates clean coding practices, modular architecture, and reliable functionality, making it suitable for real-world disaster management and future enhancements.